Layout review of the SPB leg Rev B double-pulse board

Project spb-bb-revb-dpt (KiCad 7), fab package dated 2026-10-26. Review dated 2026-10-30, prepared with Claude Code.

Table 1. Scope and basis of the review.

Item	Scope of this review
Board	SPB leg, Rev B, double-pulse test board, KTH 2026-10. Four cells of EPC2361 high-side and low-side GaN FETs with a PMK UFCS coaxial shunt in each low-side source, two Infineon 2EDF7275K isolated gate drivers, 75 V DC link, 1200 V control-to-power barrier.
Design data	Board and schematic files, design rule file, and the fab package of 2026-10-26: gerbers, drill files, position and placement files, via-in-pad list, bill of materials.
Process	JLCPCB four-layer, 2 oz copper on all layers, 0.1 mm prepreg between layers 1 and 2 and between layers 3 and 4.
Checks	KiCad 10.0.6 design rule check (DRC); geometric analysis of all copper, mask and via data extracted from the board file; regeneration of the gerber and drill files and comparison with the fab package; comparison of the custom footprints with the manufacturer drawings [1], [2], [3]; comparison of the design minima with the published JLCPCB limits [4], [5].
Conventions	Layers are named as in KiCad: F.Cu is the top layer, In1.Cu and In2.Cu are the inner layers, B.Cu is the bottom layer. Coordinates are KiCad board coordinates in mm unless marked as fab coordinates, which are measured from the auxiliary origin (fab x = x − 105.25, fab y = 160.8 − y).

1. Summary

The layout is clean and consistent with the fab package. One defect should be fixed before ordering: the switch node of each cell reaches the inner switch-node planes through a single 0.3 mm via. Everything else that was checked passed or is a minor judgement call. Section 2 describes the defect, Section 3 lists what was verified, and Section 4 lists the minor items.

2. Must fix: one via per cell on the switch node

The top-layer copper of the switch node (SW) in each cell is a small island between the high-side and low-side FET bars. The AC bus and all SW strips lie on the inner and bottom layers, and the only connection between the island and these planes is one via per cell. Table 2 lists the four islands and the single via of each.

Table 2. Top-layer SW islands and their vias. All islands extend from y = 103.1 mm to y = 111.0 mm. Every via on the board is 0.6 mm pad, 0.3 mm drill unless stated otherwise.

Cell	FETs	Island area (mm^2)	Island extent in x (mm)	Position of the single via (mm)
1	Q1, Q3	27.5	110.2 to 121.2	(120.8, 107.05)
2	Q2, Q4	27.1	131.2 to 142.2	(131.7, 107.05)
3	Q5, Q7	27.5	157.8 to 168.8	(168.3, 107.05)
4	Q6, Q8	27.1	178.8 to 189.8	(179.2, 107.05)

The full inductor current of each cell therefore passes through one plated hole. A 0.3 mm via in a 1.6 mm board has a resistance of about 1.2 mΩ, comparable to the on-resistance of the EPC2361, and it sits inside the drain-source voltage measurement path, because the MMCX centre pin taps SW on the bottom plane. The other 564 SW vias only stitch the bus and the strips to each other. By contrast, DC+ enters each cell through eight via-in-pad vias and DC− returns through about 37 vias per shunt, so the single via looks like an omission in the generator script rather than intent.

Recommendation. Add a via array to the top-layer SW zone in the generator script. In cell 1 the free copper left of the FET pair, x 110.2 to 114.7 mm and y 104.6 to 110.7 mm, offers about 19 mm^2, and the column right of the gate and Kelvin pads adds more; the other cells are mirror images. Fifteen to twenty vias of 0.6 mm pad and 0.3 mm drill per cell bring the series resistance well below 0.1 mΩ.

3. Items verified

- **DRC and fab package.** The KiCad 10 DRC reports no clearance, courtyard, mask or connectivity errors. The copper gerbers of the fab package are identical in geometry to a KiCad 10 regeneration of the current board, and the drill files match hole for hole. The silkscreen gerbers correctly subtract the mask openings; the pad flashes they contain are in clear polarity and are this subtraction, not silk over pads.
- **Connectivity.** Ignoring net names, every schematic net has the same pin set in the PCB. The only extras on the PCB are 25 test points that are not in the schematic.
- **Footprints against datasheets.** The EPC2361 copper, mask bars and the full stencil match the datasheet [1] exactly, and the pattern is solder-mask defined as EPC recommends. The 2EDF7275K pads, pitch, row spacing and pin-to-net mapping match the LGA-13 drawing and pinout [2]. The UFCS body (22 mm by 7.5 mm) and 15 mm lands match the PMK drawing [3].
- **1200 V barrier.** No control copper overlaps power copper on any pair of layers. Table 3 lists the smallest lateral distance between control-side and power-side copper on each layer. All meet the rules of the project design rule file; the smallest value, at the driver pads, is set by the package itself. All 1094 vias are tented, and the only mask-exposed via drills are the via-in-pad ones.
- **Power loop.** The commutation loop of every cell stays on the top layer from the 1 µF capacitors through both FETs and the shunt, with the DC− return plane 0.1 mm below it for its whole length. Zone connections are solid everywhere.
- **Fab files and process limits.** The position and placement files match the PCB to 0.01 mm. Hole-to-hole spacing, track width and spacing, annular rings, mask dams and the drill sizes for epoxy-filled and capped vias are all inside the published JLCPCB limits for 2 oz copper [4], [5].

Table 3. Smallest same-layer distance between control-side and power-side copper, and the applicable rule of the project design rule file.

Layer	Location of the minimum	Distance (mm)	Rule (mm)
F.Cu	Driver pads U1 and U2, between the two pin rows	3.35	package
F.Cu	Control island to driver escapes	3.83	3.0
In1.Cu	Control island to Kelvin-source planes	3.85	3.0
In2.Cu	Control island to gate-supply copper	3.15	3.0
B.Cu	Control island to SW plane	4.00	4.0

4. Minor items and judgement calls

- **Via-in-pad list is four vias short.** The DC+ vias of Table 4 sit between two same-net 0805 pads under a merged mask opening but are not in the via-in-pad list of the fab package. This is harmless at JLCPCB, which fills every via; add them to the list if another fab is used.
- **JLCPCB 2 oz inner stack-up.** JLCPCB documents 2 oz inner copper only as a 1.6 mm board with 3313 prepreg and 2 oz outer copper, and it does not publish the finished prepreg thickness for that option. The 0.1 mm loop spacing is a design parameter, so confirm it with their engineers.
- **KiCad 10 housekeeping.** 64 same-net, same-priority zone overlaps are flagged as errors in KiCad 9 and 10. They do not change the fills. A future "update PCB from schematic" would also rename every net (DCP to /DCP and so on), orphan all zones, drop the 25 PCB-only test points and overwrite 79 values. Keep using the generator scripts rather than the KiCad synchronisation.
- **DC-link spacing.** DC+ and DC− copper run 0.2 mm apart over about 80 mm on the top layer and around the REDCUBE pins on both inner layers. This meets the 75 V rule of the project but it is the floor, and a mask defect shorts the link.
- **MLCC flex risk.** Twenty 2220 X7S multilayer ceramic capacitors (MLCCs) sit within 10 mm of the M4 REDCUBE bolts. Consider soft-termination parts or a torque limit.

- **Shunt bodies.** RS1 and RS4 overhang the board edge by about 2.4 mm, and the bodies of RS2 and RS3 are 0.6 mm apart. Both shields are DC−, so contact is harmless, but check any fixture.
- **Soldering.** The REDCUBE pins and all shunt pads connect solidly into 2 oz planes on four layers, so hand soldering the shunts needs preheat.
- **Driver placement.** The shunt bodies force the drivers 15 mm from the FETs. Gate paths of about 20 mm with the Kelvin return on In1.Cu are acceptable with 0.5 Ω plus 1.5 Ω gate resistors.
- **Silkscreen.** The top silkscreen carries the text DESIGNED BY CLAUDE. Keep or remove as preferred.

Table 4. DC+ vias under merged mask openings that are missing from the via-in-pad list (fab coordinates).

Net	Fab x (mm)	Fab y (mm)	Location
DCP	14.650	58.300	0805 row of cell 1
DCP	27.350	58.300	0805 row of cell 2
DCP	62.150	58.300	0805 row of cell 3
DCP	74.850	58.300	0805 row of cell 4

References

[1] EPC, "EPC2361 datasheet", epc-co.com/epc/documents/datasheets/EPC2361_datasheet.pdf

[2] Infineon, "2EDF7275K datasheet", v02_08, infineon.com (document id 5546d462636cc8fb0163b0900069305a)

[3] PMK, "UFCS current shunt series user manual", Rev 02A, iwatsu.co.jp/wp-content/uploads/2026/02/UFCS_current-shunt-series_usermanual-Rev02A_j.pdf

[4] JLCPCB, "PCB capabilities", jlcpcb.com/capabilities/pcb-capabilities

[5] JLCPCB, "Copper weight", jlcpcb.com/help/article/jlcpcb-copper-weight